Product Brief

MOS Integrated Circuit

MC-10153

EMMA3TH

HIGHLY INTEGRATED HDTV PROCESSOR WITH MPEG4-AVC/H.264 DECODER

DESCRIPTION

The EMMA3TH is an HDTV processor supporting worldwide digital/analog broadcasting formats.
It integrates in a single chip the TV set core, which includes the CPU, AV signal processing, panel signal processing and peripheral control functions, and more. It is ideal for Full HDTV architecture.

FEATURES

- Multi-CPU system
  - Main CPU : V_R5500-compatible SS1000 core (655MIPS @ 327.68MHz)
  - Sub CPU : MIPS™ 4KEc™ core (393MIPS @ 327.68MHz)
  - Extended CPU : MIPS™ 4KEm™ core (196MIPS @ 163.84MHz)[#]
- DDR2 memory I/F (x32 + x16, Up to 4.8GByte/s)
- Stream processor supporting simultaneous de-multiplexing of 4 TSs
  - Encryption/Decryption : AES, Multi2, DVB, DES and 3DES functions
  - Supported streams : MPEG2-TS, MPEG2-PS and MPEG1, DVB, ARIB, ATSC
- Audio processor supporting two multi-audio format systems
  - MPEG BC, Dolby™ Digital, MPEG2 AAC, MPEG2 HE-AAC, Dolby™ Digital Plus
- Multi format video decoder (MPEG1, MPEG2, MPEG4 AVC/H.264, VC-1)
  - Bandwidth saving decode function
- High-performance graphics engine, 3-plane 32bpp OSD (main video: 2 planes, sub video: 1 plane)
- De-interlacer supporting Full HD
- 7x5 tap high-performance video scalar
- 14-bit signal processing picture quality adjustment controller (Picture Quality Controller) supporting xvYCC
  - LTI/CTI, 2D aperture, sharpness, APL adaptive black extension/white peak correction, contrast, hue correction, Liner – γ, MATRIX conversion, V-T correction, LCD overdrive, FRC, etc.
  - Dual Link LVDS output supporting Full HD (for main video, 1920H x 1080 V x 60 Hz, 30-bit GBR)
- Downconvert analog video output (NTSC/PAL/SECAM : YC or CVBS Output)
- 4ch high speed video ADCs with analog video switch supporting up to 1080P and WUXGA
- 3D-Comb analog video decoder with SCART support
- HDMI (Deep color and x.v.color) receiver
- Various external interfaces
  - Ethernet MAC, 2-USB2.0, PCI, SmartCard, 4-UART, 3-IIC, CSI Host, CSI Target, GIO, GPIO, etc
- Various on-chip PLLs, clock input : 32.768MHz for System
- Power supply voltage : 3.3V, 1.8V(for external interface, analog core), 1.05V(for PLL, HDMI, internal logic core)

ORDERING INFORMATION

Part Number	Package
MC-10153	933-pin plastic BGA (1.0mm pitch) (35x35)

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BLOCK DIAGRAM

